

CASE 18

IMMUNE DYSREGULATION, POLYENDOCRINOPATHY, ENTEROPATHY X-LINKED DISEASE

A failure of peripheral tolerance due to defective regulatory T cells.

The primary role of the immune system is to recognize pathogens and eliminate them from the body. However, an equally important task is to distinguish potentially dangerous antigens from those that are harmless. Countless innocuous foreign antigens are encountered every day by the lungs, gut, and skin, the interfaces between the body and the environment. Similarly, the body contains numerous self antigens that might bind to the specific antigen receptors on B and T cells. Activation of the immune system by such innocuous antigens is unnecessary and may lead to unwanted inflammation. Allergic and autoimmune diseases are well-known examples of such unwanted and potentially destructive responses.

Fortunately, unwanted immune responses are normally prevented or regulated by the phenomenon of immunologic tolerance. This is defined as nonresponsiveness of the lymphocyte population to the specific antigen, and arises at two stages of lymphocyte development. Central tolerance is the result of the removal of self-reactive lymphocytes in the central organs; an autoimmune disease due to a defect in central tolerance is described in Case 17. Peripheral tolerance, in contrast, inactivates those T and B cells that escape central tolerance and exit to the periphery. Defects in either central or peripheral tolerance can result in unwanted or excessive immune responses.

Several mechanisms of peripheral tolerance exist (Fig. 18.1). One is the network of regulatory cells that prevent or limit the activation of T cells, including self-reactive T cells, and the consequent destructive inflammatory processes. When these regulatory cells do not function properly, problems can arise. A key cell type

TOPICS BEARING
ON THIS CASE:

Peripheral tolerance

Regulatory T cells

Central tolerance

Autoimmunity

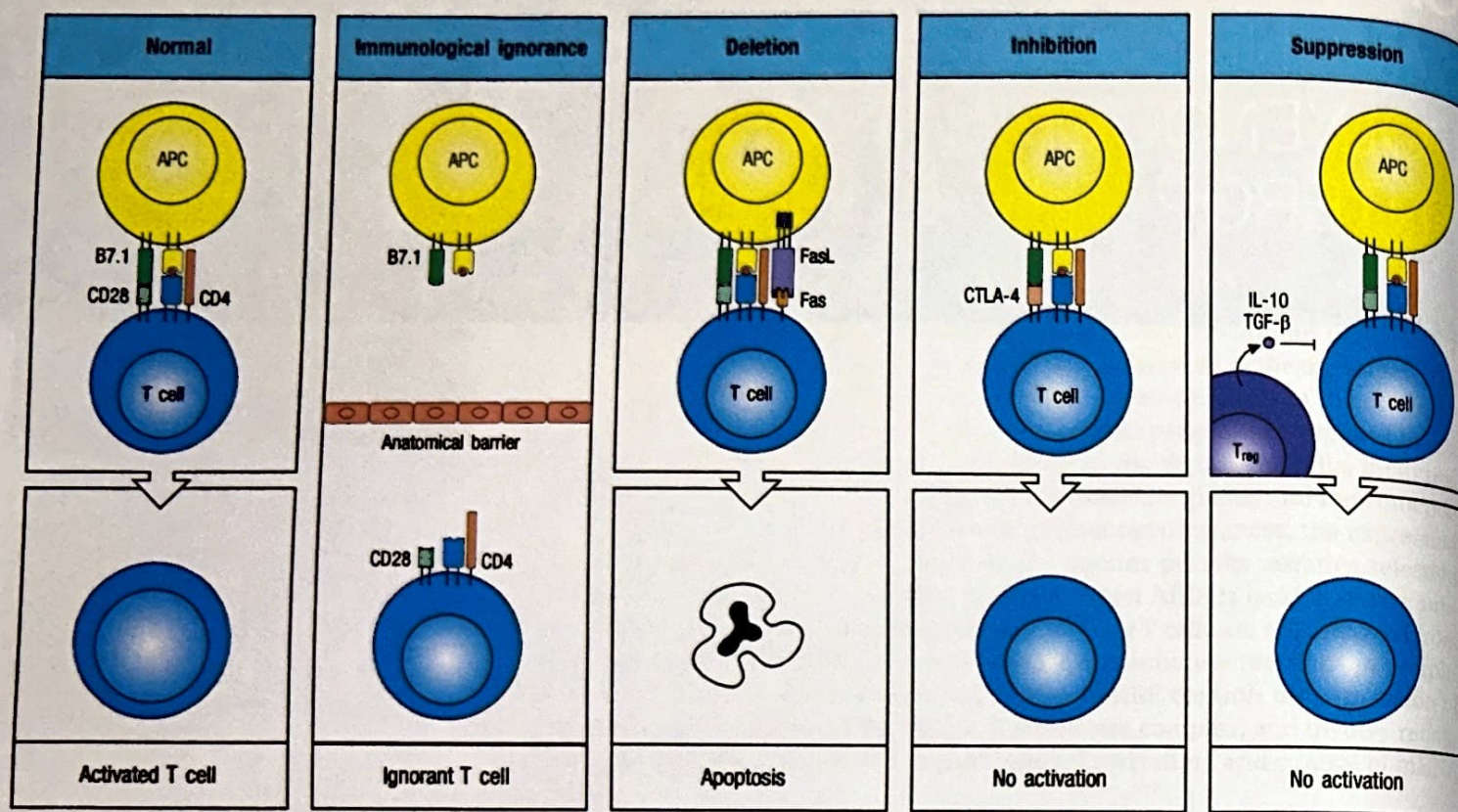


Fig. 18.1 Mechanisms of peripheral immunologic tolerance.

T cells that are physically separated from their specific antigen—for example, by the blood-brain barrier—cannot become activated, a circumstance referred to as immunological ignorance. T cells that express Fas (CD95) on their surface can receive signals from cells that express Fas ligand, leading to

their deletion. The activation of naive T cells can be inhibited if the cell-surface protein CTLA-4 (CD152) binds B7.1 (CD80) on antigen-presenting cells (APCs). Regulatory T cells (mainly CD4 CD25 Foxp3-expressing) can inhibit, or suppress, other T cells, most probably through the production of inhibitory cytokines such as IL-10 and TGF- β .

responsible for the maintenance of peripheral tolerance is the CD4 CD25 regulatory T cell (T_{reg}), also known as the natural regulatory T cell. Natural T_{reg} cells, which represent only 5% to 10% of circulating T cells, seem to become committed to a regulatory fate while still in the thymus.

Natural T_{reg} cells are CD4⁺ CD25⁺ cells that express the transcription factor Foxp3, which is essential for their specification and function as regulatory cells. Over the past decade it has emerged that these cells are crucial to the maintenance of peripheral tolerance. Neonatal thymectomy in mice and thymic hypoplasia in humans (DiGeorge syndrome; see Case 9) result in impaired generation of natural T_{reg} cells and the development of organ-specific autoimmune disease. The generation of natural T_{reg} cells in the thymus requires interaction with self-peptide:MHC class II complexes on cortical epithelial cells.

A second group of regulatory T cells seems to be induced from naive CD4 T cells in the periphery. These cells are CD4⁺ CD25⁻ and are heterogeneous, including TGF- β (TH3) and IL-10-secreting (TR1) subsets. Recently, a novel and rare population of T_{reg} cells that are CD8-positive has been described. Whereas CD4 T_{reg} subsets have been extensively studied, less is known about CD8 T_{reg} cells, their subsets, and their modes of action. NK cells and NKT cells have also been shown to be able to regulate immune responses. As a group, regulatory cells represent just one mechanism in a complex system of immunologic tolerance, acting to prevent or rein in unwanted immune responses.

The following case illustrates how a breakdown in peripheral tolerance as a result of a defect in regulatory T cells leads to a constellation of allergic symptoms, gastrointestinal symptoms, and autoimmune disease in infancy.

The case of Billy Shepherd: a defect in peripheral tolerance leading to dermatitis, diarrhea, and diabetes.

Billy was born at full term and developed atopic dermatitis shortly after birth. This was treated by skin hydration and by local application of hydrocortisone and antihistamines to control itching; the treatment was only partly successful. At 4 months of age, Billy developed an intractable watery diarrhea. Although he had initially gained weight well, by now his weight had fallen below the third centile for his age. At 6 months old, Billy started to develop high blood glucose levels and glucose in the urine. He was diagnosed with type 1 diabetes (insulin-dependent diabetes mellitus) and was referred by his pediatrician to the endocrine clinic at the Children's Hospital.

When first seen at the clinic, Billy weighed 5 kg (the third centile for age is 6.3 kg). He had diffuse eczema and sparse hair (Fig. 18.2). His cervical and axillary lymph nodes and spleen were enlarged. Laboratory tests revealed a normal white blood cell count of $7300 \mu\text{l}^{-1}$, a normal hemoglobin of 11.3 g dl^{-1} , and a normal platelet count of $435,000 \mu\text{l}^{-1}$. The percentage of eosinophils in the blood was high at 15% (normal <5%), and IgE was also elevated, at 1345 IU ml^{-1} (normal <50 IU ml^{-1}). Autoantibodies were found against glutamic acid decarboxylase (the GAD65 antigen) and against pancreatic islet cells. Billy was started on insulin therapy, which controlled his blood glucose level.

Because of the persistent diarrhea and failure to thrive, Billy required parenteral (intravenous) nutrition to maintain his weight. An endoscopy was ordered, to ascertain the cause of his persistent diarrhea, and a duodenal biopsy revealed almost total villous atrophy—an absence of villi in the lining of the duodenum—with a dense infiltrate of plasma cells and T cells (Fig. 18.3).

When Billy's mother was questioned, she revealed that there had been another son, who had died in infancy with severe diarrhea and a low platelet count. On the basis of Billy's symptoms and the family history, IPEX (Immune dysregulation, polyendocrinopathy, enteropathy X-linked) was suspected. A FACS analysis of Billy's peripheral blood mononuclear cells revealed a lack of both CD4 CD25 cells and CD4 Foxp3-positive cells. Sequencing of Billy's *FOXP3* gene revealed a missense mutation, confirming the diagnosis.

With the diagnosis established, immunosuppressive therapy, including cyclosporin and tacrolimus, was started. Billy's diarrhea, glucose control, and eczema all improved markedly. After several months, however, his symptoms began to return and he stopped gaining weight. Shortly afterwards, he developed thrombocytopenia (a deficiency of blood platelets) and anti-platelet antibodies were detected.

The decision was made for Billy to be given a bone marrow transplant from his 5-year-old HLA-identical sister. In the weeks of conditioning leading up to the transplant, the diarrhea and eczema resolved. After transplantation, full engraftment of his sister's stem cells was established. Two weeks after transplantation, anti-platelet, anti-GAD65, and anti-islet cell antibodies could not be detected. A year after the transplant, Billy continued to be symptom-free, although analysis for chimerism revealed that only 30% of his T cells are derived from his sister's cells.



Fig. 18.2 Eczematous rash on the face of a baby boy with IPEX. Photograph courtesy of Talal Chatila, UCLA.

Infant with dermatitis,
diarrhea, and diabetes.
Underlying cause?

Take family history.

Bone marrow transplant
advisable.

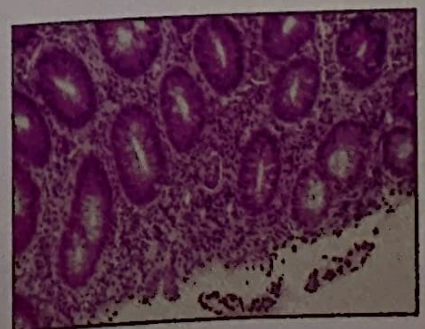


Fig. 18.3 Photomicrograph of duodenal biopsy from a child with IPEX. The section was stained with hematoxylin and eosin. Note the dense mononuclear cellular infiltrate. Photograph courtesy of Talal Chatila, UCLA.

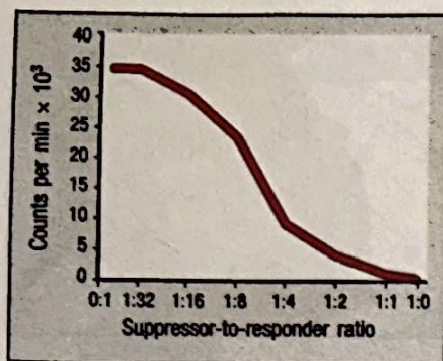


Fig. 18.4 Results of a functional assay for regulatory T cells from a normal individual. CD4 responder cells and CD4 CD25⁺ T regulatory (suppressor) cells were mixed, together with antigen-presenting cells, at the ratios shown on the horizontal axis. The cells were stimulated with immobilized plate-bound anti-CD3 and soluble anti-CD28 for 3 days, then assessed for proliferation as measured by the incorporation of ³H-labeled thymidine into DNA.

Immune dysregulation, polyendocrinopathy, enteropathy X-linked disease (IPEX).

IPEX is a very rare disease caused by mutations in the gene for the forkhead transcription factor *Foxp3*, which is essential for the function of CD4⁺ CD25⁺ T_{reg} cells. *Foxp3* expression is restricted to a small subset of TCR α : β T cells and defines two pools of regulatory T cells: CD4⁺ CD25^{high} T cells and a minor population of CD4⁺ CD25^{low/neg} T cells. Ectopic expression of *Foxp3* *in vitro* and *in vivo* is sufficient to convert naive murine CD4 T cells to T_{reg} cells. In contrast, overexpression of *FOXP3* in naive human CD4⁺ CD25⁻ T cells *in vitro* will not generate potent suppressor activity, suggesting that additional factors are required. *Foxp3* expression and suppressor function can, however, be induced in human CD4⁺ CD25⁻ *Foxp3*⁻ cells by cross-linking of the T-cell receptor and stimulation via the co-stimulatory receptor CD28, or after antigen-specific stimulation. This suggests that *de novo* generation of T_{reg} cells in the periphery may be a natural consequence of the human immune response.

T_{reg} cells are anergic *in vitro*. They fail to secrete IL-2 or proliferate in response to ligation of their T-cell receptors, and depend on the IL-2 generated by activated CD4 T cells to survive and exert their function. An *in vitro* assay that measures the ability of CD4⁺ CD25⁺ T cells to suppress CD4 T-cell proliferation is commonly used to test for T_{reg} function (Fig. 18.4). How T_{reg} cells suppress immune responses *in vivo* is still unclear. There is some evidence for contact-dependent inhibition, whereas other studies suggest that regulatory T cells exert their function by secreting immunosuppressive cytokines such as IL-10 or TGF- β , or by directly killing their target cells in a perforin-dependent manner.

Several lines of evidence show that *Foxp3* is crucial for the development and function of CD4 CD25 T_{reg} cells in mice. A mutation in *Foxp3* is responsible for an X-linked recessive inflammatory disease in the *Scurfy* mutant mouse. Male mice hemizygous for the mutation succumb to a CD4 T-cell-mediated lymphoproliferative disease characterized by wasting and multi-organ lymphocytic infiltration. T_{reg} cells are absent in *Scurfy* mice and in mice that have another spontaneous mutation in the *Foxp3* gene. In addition, specific ablation of *Foxp3* in T cells only is sufficient to induce the full lymphoproliferative autoimmune syndrome observed in the *Foxp3*-knockout mice. The *Scurfy* phenotype can be rescued by the introduction of a *Foxp3* transgene or by bone-marrow reconstitution, demonstrating the causative role of *Foxp3* in pathogenesis. Thus, the lack of *Foxp3*-expressing T_{reg} cells alone is sufficient to break self-tolerance and induce autoimmune disease.

In humans, missense or frameshift mutations and deletions in *FOXP3* result in loss of function of T_{reg} cells and uninhibited T-cell activation. As seen in Billy's case, the most common symptoms are an intractable watery diarrhea leading to failure to thrive, dermatitis, and autoimmune diabetes developing in infancy. The diarrhea is due to widespread inflammation of the gut, including the colon (colitis), that results in villous atrophy, which reduces the absorptive capacity of the intestinal lining and thus contributes to wasting. Other diseases of immune dysregulation that are seen include autoimmune thrombocytopenia, neutropenia, anemia, hepatitis, nephritis, hyperthyroidism or hypothyroidism, and food allergies. Autoantibodies also accompany these autoimmune diseases. Affected patients may also suffer more frequent infections, including sepsis, meningitis, or pneumonia, although the reason for the increased susceptibility to infection is unclear. Patients generally have normal immunoglobulin levels (except for the elevated IgE), and their ability to make specific antibody is intact.

Because the survival and function of CD4⁺ CD25⁺ T_{reg} cells depend on IL-2, which activates signaling pathways that utilize STAT5, deficiencies of either the IL-2 receptor α subunit (CD25) or STAT5B are associated with impaired T_{reg} cell function. Decreased T_{reg} function manifests as autoimmunity, food allergies, and increased

susceptibility to infections, since IL-2 is important for the expansion and function of T effector cells. STAT5B-deficient patients are of short stature, because STAT5B is required for growth hormone receptor signaling. In addition to IPEX, a number of other immunodeficiencies may affect T_{reg} function and stability, resulting in associated autoimmune manifestations. These include, for example, Wiskott-Aldrich syndrome (see Case 16), DOCK8 deficiency (see Case 50), and LRBA deficiency (see Case 53).

Questions.

- 1 Billy continued to remain asymptomatic even though he eventually only had around 30% engraftment of his sister's bone marrow stem cells (as judged from the proportions of T cells). Why is full engraftment not necessary in patients with IPEX?
- 2 Why did Billy's diarrhea improve while he was being prepared for transplantation?
- 3 Intravenous immunoglobulin (IVIG) has been used to treat IPEX. How might this be an effective therapy?
- 4 The occurrence of colitis in IPEX suggests that T_{reg} cells may be implicated in its pathogenesis and that they might be used therapeutically in the more common forms of colitis. Is there experimental data to support this claim?
- 5 What other gene mutations can give rise to a clinical picture similar to IPEX?